

4. Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
    for(int j=1; j<n; j = 2 * j)
    for(int k=1; k<n; k = k * 2)
    c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n;
```

```
scanf("%d", &n);
```

```
int c = 0;
```

```
int count = 0;
```

```
count++; // i = n/2
```

```
for (int i = n / 2; i < n; i++)
```

```
{
```

```
count++; // outer loop condition
```

```
for (int j = 1; j < n; j = 2 * j)
```

```
{
```

```
count++; // middle loop condition
```

```
for (int k = 1; k < n; k = k * 2)
```

```
{  
count++;  
c++;  
count++;  
}
```

```
count++;  
}
```

```
count++;  
}
```

```
count++;
```

```
printf("%d", count);
```

```
return 0;  
}
```

5.Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)  
{  
int rev = 0, remainder;  
while (n != 0)  
{  
remainder = n % 10;  
rev = rev * 10 + remainder;  
n/= 10;  
  
}  
print(rev);  
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf()

statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

```
#include<stdio.h>
int main()
{
int n,rev=0,remainder,counter=3;
scanf("%d",&n);
while(n!=0)
{
counter+=4;
remainder=n%10;
rev=rev*10+remainder;
n=n/10;
}
printf("%d",counter);
return 0;
}
```